

Definition. Let G be a group and let H be a nonempty subset of G . For any $a \in G$,

When H is a subgroup of G

Example 1. List all the distinct left cosets of the subgroup $K = \{R_0, V\}$ in D_4 .

Example 2. List all the distinct left cosets of the subgroup $H = \{0, 2, 4\}$ in $\mathbb{Z}_6$.

Properties of Cosets

Let H be a subgroup of G , and let a and b belong to G . Then,

1. $a \in aH$.
2. $aH = H$ if and only if $a \in H$.
3. $(ab)H = a(bH)$ and $H(ab) = (Ha)b$.
4. $aH = bH$ if and only if $a \in bH$.
5. $aH = bH$ or $aH \cap bH = \emptyset$.
6. $aH = bH$ if and only if $a^{-1}b \in H$.
7. $|aH| = |bH|$.
8. $aH = Ha$ if and only if $H = aHa^{-1}$.
9. aH is a subgroup of G if and only if $a \in H$.

Exercise 1. Prove #1.

Exercise 2. Prove #4.

Exercise 3. Prove #5.

Theorem (Lagrange's Theorem).

Proof:

Corollaries to Lagrange's Theorem:

Exercise 4. List all possible sizes for subgroups of S_4 . Is there a subgroup of every size?

Question 1. Is the converse of Lagrange's Theorem true or false? If d divides the order of a group G , then G has a subgroup of order d ?

Theorem. Let H and K be two finite subgroups of a group G .

Theorem.

Additional Exercises

1. Suppose K is a proper subgroup of H and H is a proper subgroup of G . If $|K| = 42$ and $|G| = 420$, what are the possible orders of H ? Explain.

2. Let G be a group and $|G| = 21$. If $g \in G$ and $g^{14} = e$. What are the possible orders of g ?

3. Suppose that H and K are subgroups of a group with $|H| = 24$ and $|K| = 20$. Prove that $H \cap K$ is Abelian.

4. Let G be a group of order 25. Prove that either G is cyclic or $g^5 = e$ for all $g \in G$.

5. Let H and K be subgroups of a finite group G with $H \subseteq K \subseteq G$. Prove the following:

$$|G : H| = |G : K| \cdot |K : H|$$

6. Suppose H is a subgroup of a group G , and $[G : H] = 2$. Suppose also that a and b are in G , but not in H . Prove that $ab \in H$.

7. Let G be a group with $|G| = pq$ where p and q are prime. Prove that every proper subgroup of G is cyclic.